PART 1 — SHORT SCRIPTS FOR ALL 11 WEBSITE SLIDES (pitch.html#1 to #11)

WEBSITE SLIDES #1 – #11

Matches `soulnassim.github.io/Dortmund-OFF-grid-air/pitch.html` slide by slide, followed by your Full Left-to-Right Demo

Slide #1 · Opening Hook — `Dortmund OFF-Grid Air`

pitch.html#1

🎬 **ON SCREEN: Slide #1 (3D Dortmund Hero Animation). Pause 2 seconds, smile, look at the jury.**

"Imagine ten autonomous drones flying over Dortmund right now — delivering urgent blood supplies between hospitals, or helping firefighters see a big city fire from above.

Normally, those drones rely on public 5G or the internet to stay connected. ***But what happens during a real crisis when cell towers go down, or an attacker targets the network?*** *(1s pause)*

That is exactly where our mission, Dortmund OFF-Grid Air, comes in: we have to keep all ten drones flying and connected back to HQ at the Reinoldikirche with **zero public internet and zero cell towers** — which means they have to build their own private network right in the sky!"

Slide #2 · The Problem — `Today's fleets rely on 2.4 GHz, 4G/5G & GPS. In a crisis, all three fail.`

pitch.html#2

🎬 **ON SCREEN: Slide #2. Press [→] 3 times to reveal `01 · PUBLIC SPECTRUM`, `02 · STAR TOPOLOGY`, and `03 · BLIND NAVIGATION`.**

"Why do commercial drones fail immediately in a crisis today? Because three things break at once:

• **First (`01 · PUBLIC SPECTRUM — 2.4 GHz & 4G/5G jam`):** Standard drones fly on public 2.4 GHz radio or public 4G/5G cell towers. A cheap radio jammer or power blackout wipes both out in seconds.

• **Second (`02 · STAR TOPOLOGY — No relay, no QoS`):** Every drone only talks straight to the ground and **cannot bounce signals through neighbor drones (no relay)** — so one tall building cuts it off! Even worse, there is **no traffic priority (no QoS)**, meaning heavy HD camera video clogs the weak link and chokes out tiny steering commands.

• **And third (`03 · BLIND NAVIGATION — GPS spoofed, alarms late`):** Standard autopilots blindly trust fake GPS coordinates, while standard rolling-average alarms stay blind to slow-ramping attacks for over **9 seconds**!"

Slide #3 · Our Answer — `Optical lasers beat the jammer. Now the city is the obstacle.`

pitch.html#3

🎬 **ON SCREEN: Slide #3 (U-Tower 95m diagram + `22–28 m`, `120 m`, `3 s` on the right). Press [→] 3 times.**

"**To defeat radio jammers completely, we move our primary connection onto invisible 1550 nm Free-Space Optical (FSOC) lasers!** Because a laser is a narrow beam of light — not a radio wave — a radio jammer on the ground cannot jam it!

However, lasers create a new challenge: Dortmund's 3D buildings. Look at the diagram on the left: while typical Dortmund blocks are **22 to 28 meters** tall, the **Dortmunder U-Tower reaches 95 meters**. When `DR-03` flies at **50 meters** behind the U-Tower, the building blocks the laser beam from `HQ (65 m)`!

Therefore, we place **two High-Altitude Relay Drones (`DR-01` and `DR-02`) at 120 meters** — orbiting above every roof in Dortmund — with a **3-second hysteresis** so links never flicker on building edges."

Slide #4 · Hybrid Mesh — `Two off-grid links per drone (FSOC 1550 nm + DECT NR+ 1.9 GHz)`

pitch.html#4

🎬 **ON SCREEN: Slide #4 (Two columns: FSOC Laser 1550 nm vs DECT NR+ 1.9 GHz + `POD SWaP: 225 g · 9.7 W`).**

"Every drone carries a lightweight **225-gram, 9.7-watt hybrid pod** (signed with HMAC-SHA256) running **two off-grid links that protect each other**:

• **On the left (Primary · FSOC Laser 1550 nm):** When Line-of-Sight is clear, drones talk at **1 Gigabit per second** and **2 ms latency** with **0% radio jamming loss** — though dense fog scatters laser light beyond 520 meters (Mie scattering).

• **On the right (Fallback · DECT NR+ 1.9 GHz):** ***What happens when a building or dense fog blocks the laser?*** It switches to **DECT NR+** — **the world's first license-free 5G-standard radio that lets drones build their own private network in the sky (with zero SIM cards and zero cell towers)**, diffracting around buildings and penetrating dense fog with **0 dB loss**!

• ***And what if the opposite happens — what if an attacker jams the radio?*** Then any drone with clear Line-of-Sight stays 100% immune on the green laser!"

Slide #5 · Layer 1: Jamming Physics — `Distance decides. We don't cheat it.`

pitch.html#5

🎬 **ON SCREEN: Slide #5 (Concentric circles diagram `OUTER ZONE · 620 m` & `CORE <150 m`). Press [→] 3 times.**

"Jamming power follows the inverse-square law ($1/r^2$), so distance decides what happens across three zones:

1. **Clear Line-of-Sight · Anywhere (Green):** The laser holds with **0% loss** because photons ignore radio noise.

2. **Behind a building · `>150 m` in the Outer Zone (Orange — `DR-07` at 455 m):** The building blocks the laser, so `DR-07` uses **DECT NR+** and triggers **QoS Emergency mode**: heavy video drops to **`0%`** so **C2 steering stays locked at 12 kbps (100%)**!

3. **Behind a building · `<150 m` in the Core (Purple — `DR-03` at 53 m):** Both the laser and radio are blocked at once. We don't cheat physics — the drone has to solve it physically."

PART 1 (CONT.) — WEBSITE SLIDES (#6 to #11)

Slide #6 (Pop-Up) · Slide #7 (Slow Jammer & Honest AI) · Slide #8 (LLM Co-Pilot & Fleet Learning) · Slides #9–#11

WEBSITE SLIDES #6 – #11

Slide #6 · Autonomous Recovery — `When every radio fails, climb.`

pitch.html#6

🎬 **ON SCREEN: Slide #6 (Vertical Pop-Up diagram + 4 timeline steps: `0.0 s`, `1.5 s`, `>95 m`, `3.2 s`). Press [→] 4 times.**

"***So what does a drone do when every radio fails at `0.0 s`?*** *(1s pause)* **It climbs!**

• At **`1.5 s`**: After 1.5 seconds isolated behind the U-Tower, `DR-03` switches to its onboard **internal movement and camera sensors (`INS/VIO` — needing zero GPS and zero radio)** and climbs straight up from 50 m to 125 m.

• At **`>95 m` → `3.2 s`**: The moment it clears the **95-meter U-Tower roof**, Line-of-Sight to **`Relay DR-01 (120 m)`** opens up, and in **3.2 seconds**, the green FSOC laser snaps right back online!"

Slide #7 · Layer 2: Threat Detection — `Catch the slow attack at onset (9.5s vs 0.0s)`

pitch.html#7

🎬 **ON SCREEN: Slide #7 (`Threshold rules 9.5s` vs `Contrastive prototypes 0.0s` + `Unexplained → learned`). Press [→] to reveal all items.**

1. The Slow Jammer Problem — 25 sec:

"***How will tomorrow's smart attackers bypass standard detectors?*** *(1s pause)* Instead of turning a jammer on at full power all at once, a smart attacker **increases the jamming power very slowly over time — a Ramping Jammer**. ***Why does that fool a standard detector?*** Because a standard detector only compares the current signal to the **moving average of the last 4 seconds**. When the jamming increases slowly, that 4-second average shifts down right along with the attack — just like your eyes adjusting to a slowly darkening room — so the detector stays completely blind for **9.5 seconds** (over 130 meters of flight). **Our Contrastive Detector fixes this:** at takeoff, it saves a **frozen healthy baseline** of 4 values (Signal Strength, Signal Clarity, Packet Loss, and GPS-to-INS offset). Because that takeoff baseline never shifts, we detect even a slow ramping jammer at **0.0 seconds**!"

2. The Honest AI — 15 sec:

"***Now, what if an attacker combines two weapons at once — Jamming AND GPS Spoofing — in a pattern we have never seen before?*** *(1s pause)* Most AI models blindly guess — and a wrong guess triggers the wrong defense. From my security research discussions at **MPI-SP**, **real trust comes when AI is honest about what it does NOT know**. Therefore, instead of guessing, our detector flags `DR-03` and `DR-07` in yellow as **UNEXPLAINED — needs label**."

Slide #8 · Layer 3: LLM Co-Pilot & Collaborative Fleet Learning — `The LLM never detects. It explains.`

pitch.html#8

🎬 **ON SCREEN: Slide #8 (4-step pipeline: `4D telemetry` → `Rules + contrastive detector` → `Verdict` → `LLM analyst`). Press [→] 4 times.**

3. Bridge to LLM CoPilot & Collaborative Fleet Learning — 30 sec:

"**And because this new attack needs a label, that is exactly where our LLM Security Analyst comes in!** We never let an LLM fly the drone — survival happens automatically in milliseconds, and the LLM runs **offline at HQ** strictly as a **CoPilot for the human commander**: it translates those raw sensor jumps into **two plain sentences**, opens a **critical ticket** for the ground team, and gives me this **Confirm Threat** button.

Watch what happens when I click **Confirm Threat** once: this triggers **Collaborative Fleet Learning**! Instead of every drone having to encounter this unknown attack alone, `DR-03` and `DR-07` share this tiny **4-number threat fingerprint** across the network via `HQ` — so all 10 drones immediately turn **blue**, recognize this attack in real time at 0.0 seconds, and proactively switch to their DECT backup routes to keep Recovery Time under 3 seconds!

And why is this a game-changer for clients like Morpheus Logistik? Because when one drone learns a new threat, the other nine drones don't have to abort their flights or get grounded for inspection — **drastically cutting fleet downtime, lowering operational costs, and keeping hospital deliveries on schedule!**"

Slides #9, #10 & #11 · Results (`10/10 · 3.2s · 0.0s · 225g`), Team & Launch Live Demo pitch.html#9 – #11

• **Slide #9 (Results & F&E Roadmap):** "In numbers: **10/10 drones reachable** under jamming and dense fog, **3.2s worst-case recovery**, **0.0s slow-attack detection**, and a **225g pod** plugging into Morpheus Leitstand via MAVLink 2 — scaling from Phase 1 COTS DECT NR+ bench test (Months 1–3) to Phase 2 1550 nm gimbal flight test (Months 4–8) and Phase 3 10-drone BVLOS trial between Klinikum Mitte and Westfalenhallen (Months 9–12)."

• **Slide #10 (The Team):** "Built by **Nassim Soultana** (3D Aerospace, FSOC/DECT, Fog Physics & INS/VIO Pop-Up), **Ilias Amir Soultana** (Contrastive Prototype AI, HMAC-SHA256 & LLM Co-Pilot), and **Sami Soultana** (QoS Shedding, 225g SWaP & Morpheus Ops)."

• **Slide #11 (Live Demo):** "Now let's click **Open Mission Control** and walk through the live system from left to right!"

PART 2 — FULL LEFT-TO-RIGHT LIVE DEMO SCRIPT (index.html)

LIVE DASHBOARD SCRIPT

Your Complete Left-to-Right Walkthrough of Mission Control (including the Full 3-Part Threat Intelligence & LLM Script!)

Demo Step 1 · Left Sidebar — The 10 Drones & Reciprocal FSOC + DECT NR+ Channels

Left Panel

"On the left side, you see our ten drones — `DR-01` through `DR-10` — each with its own private sky-network IP address (`10.41.0.1` to `10.41.0.10`). Every drone's goal is simply to keep a working connection back to **HQ**, our ground control base station on the roof.

To do that without public internet, our drones use **two communication channels that protect each other**:

• **First, when the sky is clear**, they connect over **green FSOC Lasers** at **1 Gigabit per second** with just 2 to 5 milliseconds of latency — fast enough to stream live HD camera video, steering commands, and telemetry straight to HQ.

• ***But what happens when a drone flies behind a tall building like the Dortmunder U-Tower — or through thick fog ([W]) and fire smoke — and the laser loses Line-of-Sight?*** *(1s pause)* Automatically, the drone switches to our second channel: **orange DECT NR+** — **the world's first license-free 5G-standard radio that lets drones build their own private network in the sky (with zero SIM cards and zero cell towers)**, penetrating buildings and thick fog, and bouncing steering commands and 3D positions from drone to drone up to our two **High-Altitude Relay Drones (`DR-01` and `DR-02`)** flying at 120 meters!

• ***And what if the opposite happens — what if an attacker blasts a radio jammer to block DECT NR+ radio?*** *(1s pause)* Because **FSOC is a narrow beam of light (not a radio wave)**, a radio jammer on the ground cannot jam the laser! Any drone with clear Line-of-Sight to HQ or our 120-meter relays stays 100% immune on the green laser."

Demo Step 2 · Center 3D Map — The 3 Attack Circles Around the U-Tower

Center 3D Map

🎬 **ACTION: Click [⚡ Jamming] and [📍 GPS Spoofing] at the bottom bar so the 3 circles appear around the U-Tower.**

"Now look at the center 3D map of Dortmund when we turn on an attack around the **95-meter Dortmunder U-Tower**. First, why are some drones like `DR-04` and `DR-08` still green? Because an attacker on the ground has a limited physical range — drones outside the circles stay untouched. Inside the attack zone, you see **three concentric rings**:

1. **In the outer Yellow Circle (GPS Spoofing):** The attacker sends fake satellite GPS coordinates. Our drone spots the sudden jump by cross-checking GPS against its own **internal movement sensor (INS)**, rejects the fake GPS, and stays on its real flight path.

2. **In the middle Red Circle (Radio Jamming — `DR-07`):** The attacker blasts heavy radio noise, so the drone switches to **DECT NR+ backup radio** and cuts off heavy video.

3. **And in the inner Purple Circle (`DOUBLE-FAIL` — `DR-03`):** Look at `DR-03` — it is flying at 50 meters right behind the 95-meter U-Tower, so the building blocks its laser **AND** the jammer crushes its radio at the same time! After 1.5 seconds isolated, the drone switches to **INS/VIO** and automatically **climbs straight up (`POP-UP` to 125m)** until it clears the 95-meter roof and catches a clean laser beam again in **3.2 seconds**!"

Demo Step 3 · Right Sidebar (Top & Middle) — Resilience Metrics & QoS Traffic Priorities

Right Top & Mid

"Now look at the top of the **Right Sidebar**: the **top four boxes (Resilience Metrics)** are our network health scorecards — showing **100% Reachability**, **Average Latency**, **Packet Loss**, and **3.2s Recovery Time**.

Right below that, look at the **four bars under Traffic Priorities (QoS)**: **C2 Command** (steering), **Telemetry** (battery and height reports), **Inter-Drone** (drones sharing 3D positions so they don't collide, aim their lasers, and route packets), and **Live Camera Video**. Because live video eats **99% of our bandwidth** while steering commands need less than **1% (12 kbps)**, our system automatically **cuts off the heavy video (`SHED · 0%`)** in an emergency — guaranteeing that **C2 Steering stays locked at 90 to 100%** so no drone ever crashes!"

Demo Step 4 · Right Sidebar (Bottom) — Threat Intelligence & LLM Security Analyst (Full 3-Part Script)

Right Bottom (70 sec)

🎬 **ACTION 1: Point to [⚡ Ramp Jammer] → ACTION 2: Click [🎯 Coordinated RF+GNSS] (Turns YELLOW) → ACTION 3: Click [✓ Confirm Threat] (Turns BLUE!).**

1. The Slow Jammer Problem ([⚡ Ramp Jammer]) — 25 sec:

"Now look at our **Threat Intelligence** box on the bottom right. ***How will tomorrow's smart attackers bypass standard detectors?*** *(1s pause)* Instead of turning a jammer on at full power all at once, a smart attacker **increases the jamming power very slowly over time — a Ramping Jammer**. ***Why does that fool a standard detector?*** Because a standard detector only compares the current signal to the **moving average of the last 4 seconds**. When the jamming increases slowly, that 4-second average shifts down right along with the attack — just like your eyes adjusting to a slowly darkening room — so the detector stays completely blind for **9.5 seconds** (over 130 meters of flight). **Our Contrastive Detector fixes this:** at takeoff, it saves a **frozen healthy baseline** of 4 values (Signal Strength, Signal Clarity, Packet Loss, and GPS-to-INS offset). Because that takeoff baseline never shifts, we detect even a slow ramping jammer at **0.0 seconds**!"

2. The Honest AI (Click [🎯 Coordinated RF+GNSS]) — 15 sec:

"***Now, what if an attacker combines two weapons at once — Jamming AND GPS Spoofing — in a pattern we have never seen before?*** *(1s pause)* Most AI models blindly guess — and a wrong guess triggers the wrong defense. From my security research discussions at **MPI-SP**, **real trust comes when AI is honest about what it does NOT know**. Therefore, instead of guessing, our detector flags `DR-03` and `DR-07` in yellow as **UNEXPLAINED — needs label**."

3. Bridge to LLM CoPilot & Collaborative Fleet Learning (Click [✓ Confirm Threat]) — 30 sec:

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Technical Deep-Dive Notes (For Jury Q&A Only)

How Z-Score, Cosine Similarity, Detector-to-LLM JSON, and Hardware Scope work under the hood

Q&A READY

1. How Z-Score (>= 4.5) and Cosine Similarity (>= 0.84) Work Under the Hood

• **Step 1 — Anomaly Detection (Z-Score):** During the 40s (80-tick) takeoff warmup, each drone records the mean (μ) and standard deviation (σ) of 4 sensors: `[RSSI, SNR, Packet_Loss, GNSS_INS_Offset]`. Every 0.5s tick, we compute the 4D Z-score deviation vector `[z_rssi, z_snr, z_loss, z_gnss]`. If its Euclidean length `z_score >= 4.5` (e.g., `z = 24.6`), an attack is detected at `0.0s`.

• **Step 2 — Attribution (Cosine Similarity):** We compare the *direction* of that 4D Z-vector against known attack prototypes using Cosine Similarity (`sim` from `0.00` to `1.00`). Pure `jamming` points along `[High, High, Med, 0]`; pure `spoofing` points along `[0, 0, 0, High]`. When both hit together (`Coordinated RF+GNSS`), the vector points diagonally between them: max similarity is only `sim = 0.68 < 0.84` threshold → flagged as `UNEXPLAINED`.

2. How the Detector Feeds the LLM & How 1-Click Collaborative Learning Works

• **Detector → LLM JSON:** The detector passes a structured JSON payload `{device: "DR-07", label: "UNEXPLAINED", z_score: 24.6, dims: {rssi: 8.1, snr: 18.2, loss: 4.0, gnss_offset_m: 16.4}, similarity: 0.68}` to the LLM Analyst (`llm_analyst.py`), which runs offline at HQ (with optional live API mode to `gpt-6-studierende` on `chat.kiconnect.nrw/v1/responses`).

• **1-Click Fleet Learning (`CONFIRM_THREAT`):** Clicking Confirm Threat normalizes the 4D Z-vector into a unit vector (4 floats = 16 bytes) and saves it in the shared fleet prototype table as `coordinated_rf_gnss`. On the very next tick, similarity jumps from `0.68` to `0.99 >= 0.84` across all 10 drones.

3. Why Only 4 Sensors in the Detector? & How Do Moving Drones Aim Lasers?

• **Why we excluded Latency & Velocity from the 4D detector (Invariance):** When a drone flies behind the U-Tower and switches from FSOC Laser to DECT radio, its latency jumps from 5ms to 40ms. If latency or speed were inside the cyberattack detector, every building in Dortmund would trigger a false cyberattack alarm! Keeping only link-budget & navigation residuals gives us **0 false alarms** across 1,200 noisy drone-ticks.

• **How drones aim FSOC lasers (PAT) & Hardware Economics:** Drones broadcast their 3D GPS/INS coordinates over omnidirectional **DECT NR+** (`Inter-Drone` channel) so gimbals know where to aim the laser. And because our MEMS optical gimbal weighs 155g and our Nordic nRF9151 DECT NR+ radio weighs 70g (total **225g / 9.7W** hybrid pod), our architecture puts high-altitude relays on `DR-01` and `DR-02` at 120m while allowing lower drones to use DECT NR+ seamlessly.